Due: August 21, 2026

1. **Integration by Substitution (u-substitution):** Calculate the following indefinite and definite integrals:

   a) $\int 2x(x^2 + 4)^5 dx$

   b) $\int xe^{x^2} dx$

   c) $\int_0^1 x\sqrt{1 - x^2} dx$

2. **Integration by Parts:** Calculate the following integrals using integration by parts ($\int udv = uv - \int vdu$):

   a) $\int xe^x dx$

   b) $\int x \ln(x) dx$

   c) $\int_1^e \ln(x) dx$

3. **Improper Integrals:** Evaluate the following improper integrals, determining whether they converge or diverge:

   a) $\int_1^\infty \frac{1}{x^2} dx$

   b) $\int_0^\infty e^{-x} dx$

   c) $\int_1^\infty \frac{1}{x} dx$

4. **Double Integrals:** Evaluate the following double integral over the specified rectangular region:

$$\int_0^1 \int_1^2 (2x + 6y) dy dx$$

5. **R Basics Questions:**

   a) Suppose you have a vector ‘x <- c(1, 3, 5, 7, 9)‘. Write the R code to:

      i. Compute the mean of ‘x‘.

      ii. Extract the third element of ‘x‘.

      iii. Find all elements in ‘x‘ that are greater than 4.

   b) In R, what is the difference between a list and a data frame? Which would you use to store a spreadsheet of country-level GDP and population data?

6. **Bonus Question (Linear Regression):** We have talked a lot about linear regression in the last two weeks, and we know we can compute regression coefficients with $\beta = (X^T X)^{-1} X^T Y$. Given that X is the left-hand side matrix and Y is the right-hand side vector below, find β. Then use elementary row operations to solve the augmented matrix as given. What do you notice about the results from the regression and those from solving the linear system?

$$\left[\begin{array}{cc|c} 3 & 4 & 2 \\ 1 & 1 & 1 \end{array}\right]$$